IDENTIFICATION OF GROUNDWATER POTENTIAL ZONES USING REMOTE SENSING AND GIS: A CASE STUDY FROM KURNOOL DISTRICT, ANDHRA PRADESH, INDIA

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ABSTRACT

Water depletion due to growing urbanisation and population in urban and rural areas needs artificial groundwater recharge preparation. Remote Sensing (RS), Geographic Information System (GIS) and Analytical Hierarchy Approaches (AHP) are advantageous methods for detailing possible groundwater potential zones (GWPZ) in arid and semi-arid India. An array of eight sets of thematic layer criteria/factors influences the future groundwater storage of the investigation field created the region's groundwater capacity field. The analysis shows that 2.30 km^2 has rather high groundwater capacity, 162.10 km^2 has GWP, while 127.78 km^2 and 1.45 km^2 below moderate and low GWPZ. The results of the examination of the effect of criteria discernment revealed that the limit of the procedure to convey precise figure is necessary to the extent of the criteria procedure used. It was established in the examination that the GIS-based basic AHP methodology is fit to create precise and reliable estimates, especially if the criteria used for desire are lucid. Such findings showed that in areas of knowledge scarcity via the AHP methodology, developers will likely have accurate groundwater resource learning based on geospatial information research. Thus, the generating condition for potential groundwater mining arrangement may be done skilfully.

KEYWORDS: Groundwater, AHP, Remote Sensing, GIS, Exploration.

INTRODUCTION

Groundwater is a prime hotspot for drinking and irrigation system, anyway this fundamental asset is effectively at risk for contamination because of human tendency to devour a greater amount of the groundwater [29]& [3]. The quick development of urban zones has influenced the groundwater quality due to over misuse of assets and ill-advised waste transfer rehearses. The nature of groundwater fluctuates from place to place with the profundity of water table [10]& [9]. It may also vary with the seasonal changes. Therefore, in many groundwater studies, evaluation of the quality of groundwater important as the quantity [19]. Water quality is controlled by the solutes and gases broke up in the water, just as the issue suspended in and gliding on the water [25].

It is the culmination of normal physical and chemical water state, much as other changes that might have occurred as a result of human activity. RS and GIS are commonly used to discover and control various natural resources, i.e. freshwater, rocks, etc. The approximate water level elevation was effectively established in the current years using geospatial techniques [21]. Remote Sensing method provides leeway to massive participation, including in inaccessible regions. It provides profitable knowledge on the like geology, lineament density, lineaments, slope, geomorphology, drainage density, soil, rainfall and land use / land cover (LULC) easily and cost-effectively [5] & [26]. A combined GWPZ map has been categorised as bad, moderate, fine, and very good based on groundwater accessibility in the study area, etc. In semi-arid regions (e.g. Andhra Pradesh: Anantapur, Kadapa) surrounded by hard rock terrain, groundwater can be found in shallow aquifers that floor the major valleys and fractured areas in the bedrocks [4], [13] & [19].

The water-bearing formations in the intermountain basin store vast proportions of non-inexhaustible that are known to have been recharged in past rainy climatic times. Aquifers in the shallow fracture region provide sustainable water recharged by direct infiltration from monsoonal runoff [8], [16], [18] & [31]. Ground waters aggregations can be contained in crystalline rocks in twisted areas such as shear and fault zones. Choosing the fracture framework scheme or predicting when groundwater would likely occur in fractures is challenging. Drilling is the most ideal way to prove groundwater proximity [4].

In the present years, the importance of coupling geospatial approaches in GWP assessments has been recognised by various scientists as the outline and differentiation of groundwater properties utilising geospatial knowledge depends on a circuitous analysis of clearly perceptible surface environments such as geomorphology, geology, slope, LULC and hydrology [21].With geospatial information's skills, different databases can be combined to construct computational model for GWPZ recognition and estimation. An attempt was made to represent GWPZs in Andhra

Pradesh, India's Kurnool. With this intention, Yemmiganur Mandal is chosen with the goals to (1) understand the different thematic layer properties on groundwater (2) demarcate possible groundwater sites (3) assess the determined site with field overview to test the accuracy of the technique adopted in the analysis.

STUDY AREA

The study area covers an area of 296.6 km^2 from 15°36'0" to 15° 54'0" N latitudes and 77° 28' 30" to 77° 37' 30" E longitudes (Fig 1). The research region consists of lithological units such as granites, granite gneiss, and Meta andesites. Much of the research region consisted of granites combining granite gneisses with meta-andesite formation. The study receives an annual rainfall is around 731.09mm. The region has numerous geomorphological characteristics such as anthropogenic landscape, Denudational sources-poor dissected hills and valleys, structural origin with low dissected hills, pediment-pediplain complex with water bodies. Available water is sufficient throughout the early part of the winter season; later the water level declines rapidly, and by summer a significant number of shallow wells hitting the upper unconfined aquifers either ended up dry or did not reach the necessary. In these lines, there is an extraordinary water insufficiency in summer, including the fact there's a lot of surface water. Thus, areas need to be recognised where straightforward artificial recharge strategies may be grasped to extend water storage.

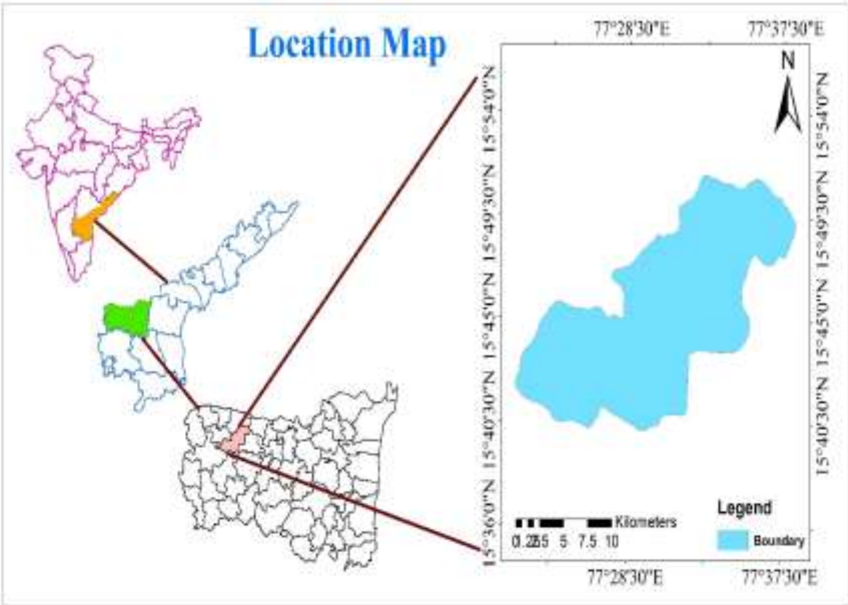

Figure.1: Location map.

MATERIALS AND METHODS

Thematic Layers

GIS was used as a primary tool in this research to create numerous thematic maps of the study region. Geology, DD (Drainage Density), LD (Lineament Density), and GM (Geomorphology) thematic layers were created using satellite imagery and SOI maps [14] & [34]. The slope map was created using the ArcGIS environment and the Cartosat Digital Elevation Model (DEM). The rainfall interpolation map was created using collateral data collected from the Chief Planning Officer at the District Collector Office. All the thematic layers are combined into an ArcGIS environment, assisting in the identification of GWPZs in the study area (Fig 2).

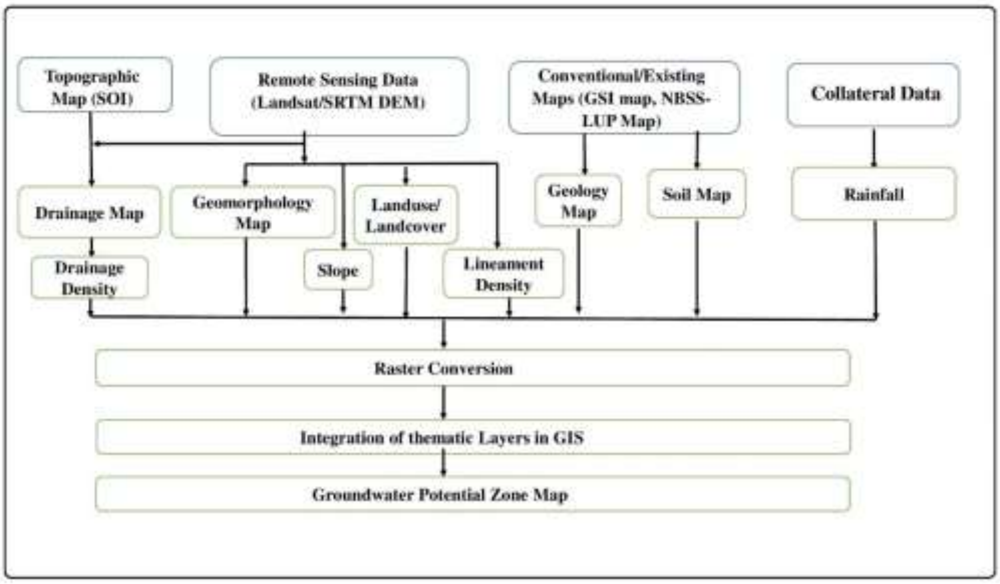

Fig.2: Methodology for delineation of groundwater potential zones of the study area.

Validation of GWPZ map

The current study's GWPZ map was validated using available well yield data from the Groundwater Department of Kurnool, Andhra Pradesh, as well as field verification to double-check the accuracy of the current work in the various GWPZs.

RESULTS AND DISCUSSION

Lithology

Geological layer prepared from the Geological Survey of India (GSI) as reference maps. The present study secured by with a major part as granites and some parts with granite gneiss and Meta andesites (Fig. 3) (GSI, 2002). A secluded patch of meta-andesites forms in the northern part of the town. The importance of aquifer form to groundwater conservation was determined using the pair wise comparison matrix of the geology diagram. In the study area, different Lithology were graded based on their properties on aquifer form having higher fractures in granites of the study area.

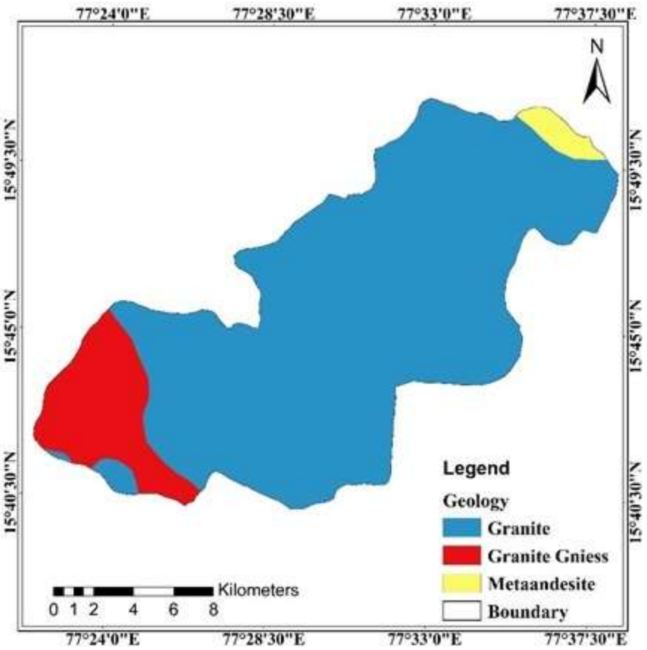

Figure 3: Geology

Geomorphology

Geomorphological maps illustrate fundamental landforms and aid in the comprehension of groundwater storage techniques, facilities, and topographical controls, as well as groundwater possibilities. These maps illustrate GWPZ's identification. Since the availability of early Landsat data, satellite images have been widely used in geomorphologic mapping. Although the primary focus has been on landform grouping, process representation, and the relationship between landforms and processes, RS is also capable of providing data on the area and dispersion of landforms, surface-subsurface composition, and topographical elevation [32].The present study established an anthropogenic origin through anthropogenic terrain and valleys, as well as a pediment-pediplain complex with low dissected hills (Fig 4).

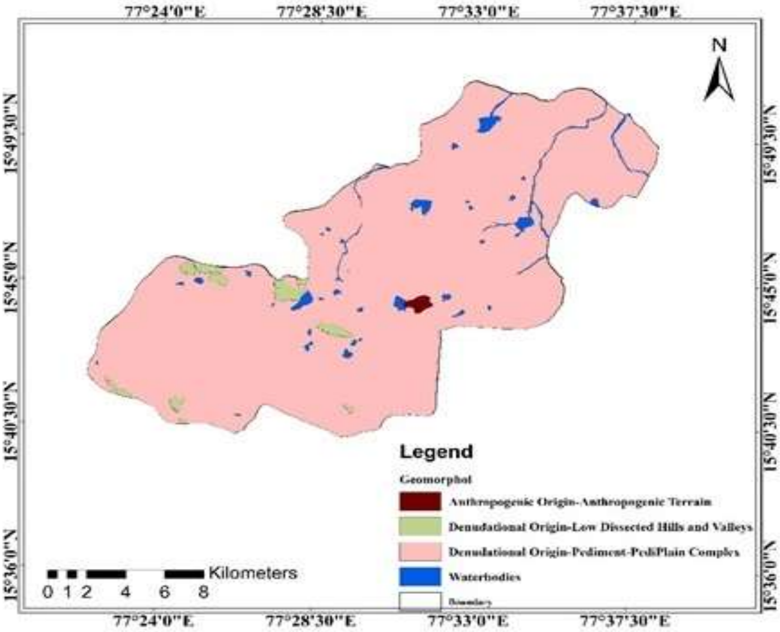

Figure 4: Geomorphology

Lineament Density

The proximity of lineaments can act as a channel for groundwater exploration, resulting in increased auxiliary porosity and thus filling in as a groundwater potential region. The Lineament Density (LD) map represents a percentage of the computed duration of the direct dimension in the gird [29].The LD of an area implicitly reveals the territory's groundwater potential, as the proximity of lineaments generally indicates a pervious sector. Areas with a higher LD value are advantageous for groundwater potentials. LD changes from 2.23 to 2.92 km/km^2 in most of the sample zone, which is restricted to the area (Fig. 5).

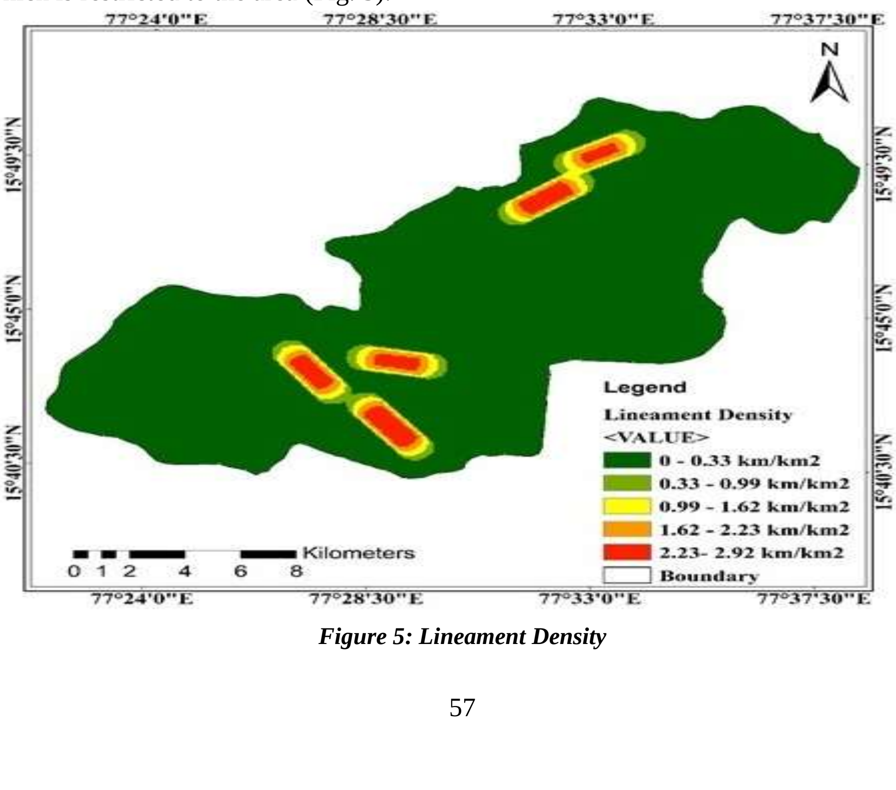

Figure 5: Lineament Density

Drainage Density

Drainage density (DD) is a metric that indicates the extent to which a basin's drainage system has developed. It represents the near proximity of channel spacing, which is caused by differential weathering of various formations, relaxation, and rainfall in a region. Horton (1932) defines the DD as the length of stream per unit of drainage area separated by the drainage basins' area [7]. Low DD is favoured in areas with extremely permeable subsoil materials, followed by thick DD in areas with impermeable subsoil materials, sparse foliage, and mountain relief. In the study region, the DD calculated using Horton's method is approximately 9.97 km/km^2 (Fig. 6). It implies that the DD is poor, indicating the presence of a permeable subsurface stratum with thick vegetation and low relief, which is characteristic of coarse DD.

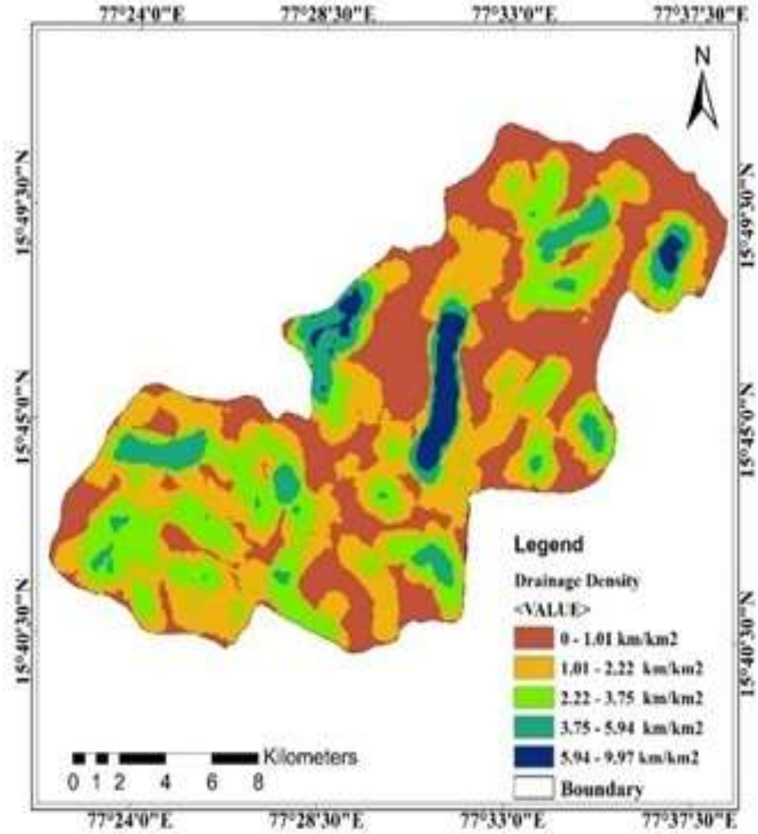

Figure 6: Drainage Density

Slope

The slope is a quantity that expresses the transition in surface values over time which may be represented in degrees or as a percentage. A raster in the DEM format is a grid in which each cell represents a value in relation to a common reference point. Since measuring the angle, it is possible to determine the maximum deviation and slope. The findings of the maps are used to construct a diagram depicting how biased neighbourhoods using ArcGIS 10.4 Spatial Analyst tools [27]. The present study's slope chart is divided into four categories: 0–3^0 (Gentle), 3-6^0 (Moderate), 6-15^0 (Steep), and 15–35^{0}' >35^0 (Very Steep) (Fig. 7). It is discovered that a considerable portion of the area is below a gentle and moderate hill, implying that the zone has a virtually flat topography. The slope is a critical parameter without which every landscape's runoff and intrusion will be regulated by delay. Runoff in regions with a more favourable slope results in less intrusion. This section is primarily responsible for the creation of aquifers.

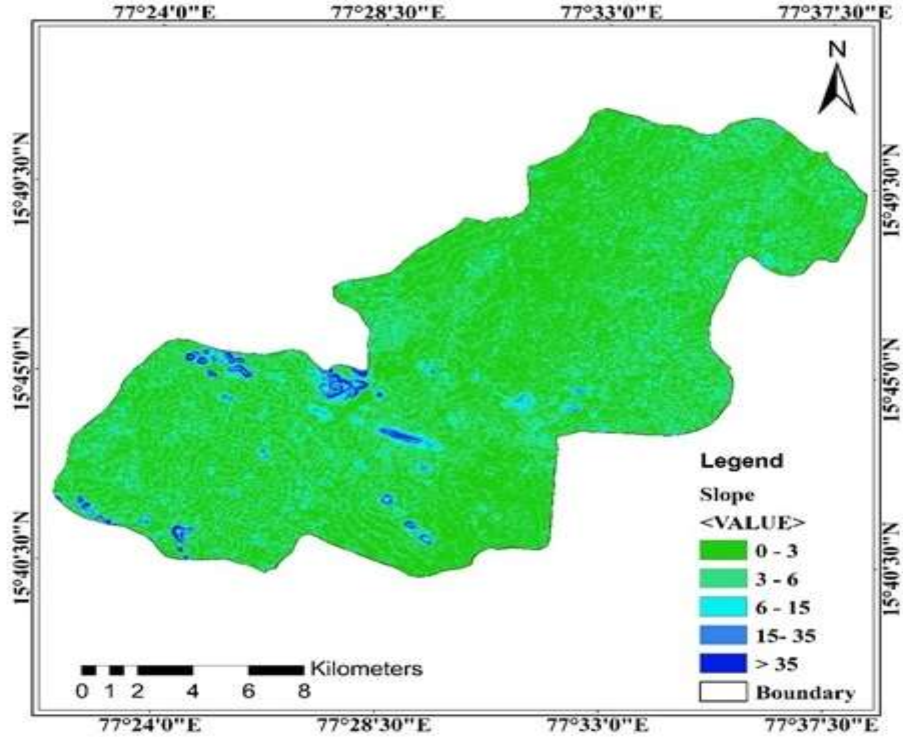

Figure 7: Slope

Rainfall

The total annual rainfall in the current report is about 731.09 millimetres in the years 2016-18. Fig 8 depicts the current study's rainfall map. The current research is mostly dependent on the north-east monsoon [16] & [22]. The north-east portion of this sample receives approximately 529.36 millimetres of precipitation each year, while the eastern portion receives approximately 356.46–441.11 millimetres per year. Precipitation is estimated as being about 731.09 mm/year in the southern portion and as being very poor in the north-east part. Precipitation dispersion along the slope gradient in the upstream south-west section has a direct effect on the infiltration rate, thus increasing the risk of GWPZs in the downstream northern section.

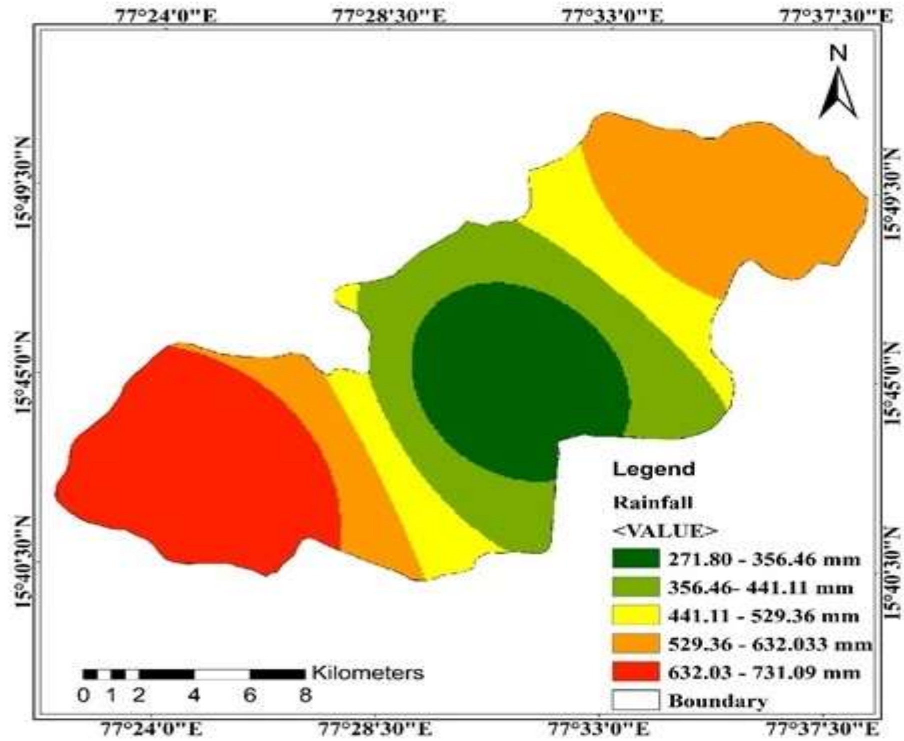

Figure 8: Rainfall

Groundwater potential zones (GWPZs)

The GWPZs were analysed using a normalised weighting of the distinct features of the thematic charts (Table 3). The GWPZ is classified into four categories: bad, moderate, decent, and very strong (Fig. 9). The loamy calcareous soils in the research area's north-west were classified as a moderate to good GWPZ. The research shows that only 4.90 km2 (1.67 percent) of the present study has a good GWP, while 19.92 km2 (58.37 percent) has a good GWP. But 78.25 km2 (229.24 percent) and 0.15 (0.45 percent) of the present study, respectively, have a moderate or low GWPZ. A careful examination of the GWPZ chart reveals that the arrangement of groundwater is almost identical to the precipitation and geographical formations next to the ridge [9] & [28].

Ground Water Potential Zone	Percentage of the area	Area (km2)
Poor	0.15	0.45
Moderate	78.25	229.24
Good	19.92	58.37
Very Good	1.67	4.90

Table 1Groundwater potential zones

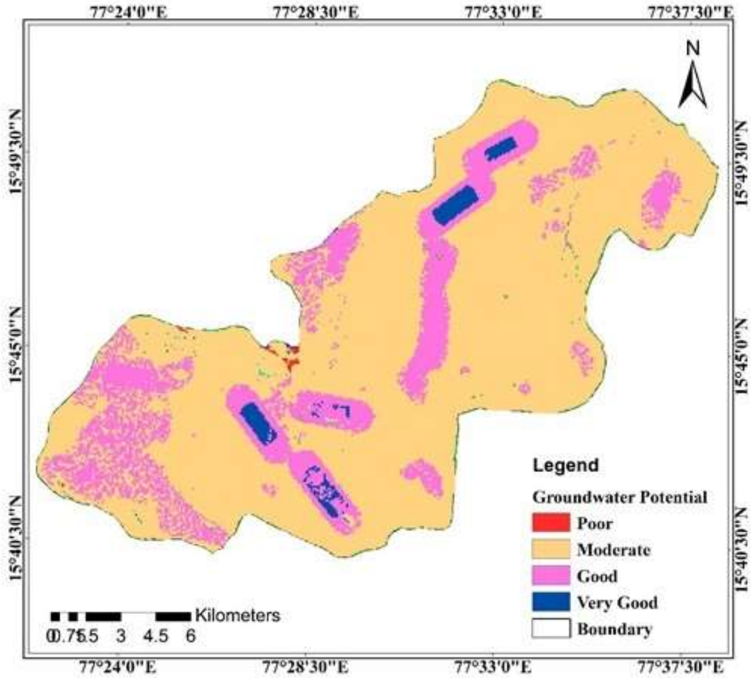

Figure 9: Groundwater Potential Zones

CONCLUSIONS

The detailed efficiency of the AHP technique in conjunction with GIS enables the implementation of an accurate and reliable spatial data processing methodology. Integrating and analysing several thematic datasets aided in the delineation of GWPZ. The aim of this analysis is to describe the GWPZ presences by dissecting distinctive thematic maps as potential influencing factors. The research reveals that only 4.90 km^2 (1.67 percent) of the study area has a really strong GWP, whereas 19.92% (58.37 km^2) has a good GWP. But, 78.25% (229.24 km^2) and 0.15% (0.45 km^2) of the study area have a reasonable or low GWP. The good GWP is determined by the Lithology type, which includes weathered rock formations and the lower slope within the study areas downstream. The GWPZ map was validated against well yield data to determine its validity, and the results were found to be consistent with the well yield data. Groundwater pervasiveness is limited in this analysis by DD, geomorphology, geology, slope, and LULC from GIS and direct perception in the region. This GWP map would be used to advise municipal governments and developers regarding areas appropriate for prospective groundwater research and to safeguard the environment from pollution caused by domestic and industrial activities.

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